Application Materials

PEARL Lab / ERC Starting Grant SIREN

Applicant: Jiajie Zhang

ShanghaiTech University

Master of Science in Computer Science and Technology

Target: PEARL Lab, Technical University of Darmstadt

Project: Structured Interactive Perception and Learning for Holistic Robotic Embodied Intelligence (SIREN)

Contents

1. **Detailed CV**
2. **Motivation Letter**
3. **Research Statement**
4. **Degree Certificates and Academic Transcripts**
   - 4.1 Master's Enrollment Certificate
   - 4.2 Master's Academic Transcript
   - 4.3 Bachelor's Degree Certificate
   - 4.4 Bachelor's Academic Transcript
5. **Academic References**

Jiajie Zhang

Mobile Autonomous Robotic Systems(MARS) Lab , ShanghaiTech University

jerryzhangamen7@gmail.com | (+86) 19858214897 | Shanghai, China |

https://jiajiezhang7.github.io/

Experience

AI R&D Center, Central Research Institute, Wolong Electric — Shanghai, China
Research Assistant — July 2025 - Oct. 2025

- Supervised by Alexander Kleiner

ShanghaiTech University — Shanghai, China
Teaching Assistant — Sept. 2024 - Jan. 2025
Mobile Manipulation
https://robotics.shanghaitech.edu.cn/teaching/moma2024

Education

ShanghaiTech University — Master of Science • 3.51/4.0 GPA
Computer Science and Technology — Sep 2023 - Present

- Selected Courses: Robotics, Mobile Manipulation, Deep Learning
- Supervised by Professor Soeren Schwertfeger

Zhengzhou University — Bachelor of Engineering • 3.66/4.0 GPA
Automation — Sep 2019 - Jun. 2023

Selected Publications

From Observation to Action: Latent Action-based Primitive Segmentation for VLA Pre-training in Industrial Settings
Accepted By CVPR 2026
Jiajie Zhang†, Sören Schwertfeger, and Alexander Kleiner
https://jiajiezhang7.github.io/latent-action-primitive-segmenter/

Generation of Indoor Open Street Maps for Robot Navigation from CAD Files
Under review for IROS 2026
JiajieZhang†, ShenruiWu, Xu Ma and Sören Schwertfeger

https://arxiv.org/abs/2507.00552

osmAG-Nav: A Hierarchical Semantic Topometric Navigation Stack for Robust Lifelong Indoor Autonomy
Under review for Autonomous Robots
Yongqi Zhang†, **Jiajie Zhang†**, Chengqian Li, Fujing Xie and Sören Schwertfeger
https://jiajiezhang7.github.io/osmAG-Nav/

Intelligent LiDAR Navigation: Leveraging External Information and Semantic Maps with LLM as Copilot
Accepted by IROS 2025
Fujing Xie†, **Jiajie Zhang** and Sören Schwertfeger

https://arxiv.org/abs/2409.08493

Neural Surfel Reconstruction: Addressing Loop Closure Challenges in Large-Scale 3D Neural Scene Mapping
Sensors (Basel, Switzerland)
Jiadi Cui†, **Jiajie Zhang†**, Laurent Kneip and Sören Schwertfeger

https://www.mdpi.com/1424-8220/24/21/6919

Projects

AGLoc++: WiFi-Fused Global Localization and Monte Carlo Enhanced Tracking in Hierarchical Area Graph Nov. 2024 - Present

- **Led AGLoc++ development**: Ported Area Graph LiDAR localization system to ROS2, integrated with the Nav2 stack, implemented WiFi-aided kidnap recovery, developed odometry-fused Monte Carlo tracking with advanced re-localization.
- **Enhance Robustness:** implement techniques such as clutter filtering, architectural matching, weighted ICP, corridor-aware downsampling,ensuring robust, high-precision localization in dynamic indoor environments.

https://jiajiezhang7.github.io/projects/1_agloc_project/

Campus Autonomy: Building and Navigating Autonomous Robots with Navigation2 Sept. 2024 - Jan. 2025

- Integrated advanced hardware, including the Agile X HUNTER SE **Ackermann** robot, **Hesai PandarQT64** Lidar, and **Insta360 Air** panoramic camera etc.
- Integrated navigation functionalities using the **ROS2** framework and **Navigation2** package, with testing performed in Gazebo simulator and on physical robots.
- Specialized in motion planning and control, applying and evaluating global planning algorithms (**Dijkstra, Hybrid A*, RRT***) for pathfinding within the Nav2 stack.

https://jiajiezhang7.github.io/projects/2_campus_autonomy/

SLAM with Vertical Plane Segmentation for Lifelong Indoor Mapping Mar. 2024 - Jul. 2024

- Integrated ROS1 Noetic with **PCL** and **Gmapping**
- Developed a RANSAC-based algorithm to extract permanent vertical structures (e.g., walls) from 3D LiDAR point clouds, filtering out temporary obstacles to produce clean, long-lasting indoor maps

https://jiajiezhang7.github.io/projects/3_slam_project/

HRI Voice Interaction Robot: 360° Perception and Proactive Dialogue on ROS2 Feb. 2025- May 2025

- **Developed an end-to-end HRI pipeline** on **ROS2 Iron**, integrating **360° fisheye perception**, **YOLOv8**, and **MTCNN** for robust face detection and servo-driven gaze tracking to maintain eye contact.
- **Engineered proactive dialogue capabilities** using **Volcano Engine LLM** for reasoning and Bilingual ASR/TTS (Baidu/StepFun), achieving **<3s latency** and natural intent analysis for elevator service scenarios.
- **Designed a modular state machine** to manage interaction flows (Idle to Decision), achieving **>95% detection accuracy** and enabling stable, contactless assistance in dynamic public environments.

https://jiajiezhang7.github.io/projects/6_hri_project/

Skills

Python
Object-Oriented Programming (OOP), Data Structures, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Matplotlib

C++
Object-Oriented Design (OOD), Encapsulation, Inheritance, Polymorphism, Smart Pointers, Memory Management, STL.

Tools
ROS/ROS2, Navigation2, PyTorch, Linux, Git, LaTex, Cursor, Windsurf

Languages

English
TOEFL iBT: 106 (Reading: 30/30, Writing: 27/30, Listening: 27/30, Speaking: 22/30)

2. Motivation Letter

Tailored to Prof. Georgia Chalvatzaki, PEARL Lab, and the SIREN project

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Dear Professor Chalvatzaki,

I am writing to apply for a PhD position in your PEARL Lab starting Fall 2026, with a strong desire to contribute to the ERC SIREN project. Your vision of coupling the robot and its environment into a unified system through structured interactive perception and learning feels like the natural next step for the kind of Embodied AI I have been trying to build during my Master's at ShanghaiTech University.

For the past two years in the MARS Lab, I have focused on developing "System-level Scaffolding" – using classical robotics principles as physical inductive biases to make foundation models more reliable and sample-efficient in real-world settings. Early on, I realized that safe point-to-point navigation is no longer the main challenge. The real difficulty lies in long-horizon mobile manipulation, where robots must actively interact with dynamic, unstructured environments over extended periods. This realization has shaped everything I have worked on since.

In my CVPR 2026 paper (first author), I introduced an unsupervised framework that extracts latent action primitives directly from unstructured industrial videos. By defining "latent action energy," the method segments meaningful motion primitives without labels or task supervision. Done in collaboration with Dr. Alexander Kleiner during my research assistantship at Wolong Electric AI R&D Center, this work turns raw observation streams into structured, reusable action representations – exactly the kind of grounding and actionable robot-environment modeling that SIREN aims to achieve in its multisensory representation learning direction.

I further tested this philosophy in practice through an LLM-Copilot navigation system (IROS 2025), which combines high-level LLM reasoning with a robust ROS2/Nav2 stack via our semantic topometric map (osmAG). I then extended osmAG into a hierarchical structure (IROS 2026 submission) to support lifelong localization and reasoning even when maps become outdated. These projects have given me solid hands-on experience in Sim2Real transfer, sensor fusion, ROS2 deployment, and real-robot troubleshooting.

What draws me most to SIREN is the opportunity to push these ideas further. My latent action primitives and physical scaffolding can strengthen the structured VLA components and physics-aware dynamics models in SIREN, while osmAG's semantic priors can help build more coherent, adaptable world representations. I am especially excited about exploring how these structured action representations can feed into action-conditioned world models for safer long-horizon planning and human-in-the-loop adaptation.

I am eager to bring my systems-oriented mindset and industrial deployment experience to PEARL Lab. Under your guidance, I believe I can help translate these concepts into real robotic systems that operate reliably in open environments. Thank you for considering my application. I would be thrilled to discuss how my background could support the SIREN project.

Best regards,

Jiajie Zhang
Master's Student, ShanghaiTech University
https://jiajiezhang7.github.io/

3. Research Statement

Fit with PEARL Lab and the SIREN project

I am applying to the PEARL Lab because your ERC SIREN project represents exactly the kind of structured, interactive approach to Embodied AI that I have been pursuing through my Master's work. SIREN's goal of coupling robot and environment into a unified system — through grounded perception, structured world models, and long-horizon interactive learning — resonates deeply with my belief that foundation models alone are not enough; they need strong physical scaffolding to become reliable in unstructured real-world settings.

My research centers on building "System-level Scaffolding" for Embodied AI: using classical robotics principles (kinematics, control, and physical constraints) as inductive biases to guide foundation models toward sample-efficient and deployable solutions. Coming from a mobile navigation background in the MARS Lab, I realized early that safe point-to-point movement is no longer the primary bottleneck. The real challenge is enabling robots to actively interact with dynamic, open environments over long horizons. This realization has driven my focus on long-horizon mobile manipulation.

In my CVPR 2026 paper (first author), I developed an unsupervised framework that extracts latent action primitives directly from unstructured industrial videos. By introducing the concept of "latent action energy," the method segments meaningful motion primitives without any labels or task-specific supervision. This work, conducted in collaboration with Dr. Alexander Kleiner during my research assistantship at Wolong Electric AI R&D Center, provides a concrete form of physical scaffolding that constrains the action space of Vision-Language-Action (VLA) models. It turns raw observation streams into structured, reusable action representations — precisely the kind of grounding mechanism SIREN aims to achieve.

Building on this, I created an LLM-Copilot navigation system (IROS 2025) that integrates high-level LLM reasoning with a robust ROS2/Nav2 stack through our semantic topometric map (osmAG). I further extended osmAG into a hierarchical structure (IROS 2026 submission) to support lifelong localization and reasoning even when maps become obsolete. These projects have given me hands-on expertise in Sim2Real transfer, sensor fusion, ROS2 deployment, and real-robot troubleshooting on both mobile bases and manipulation platforms.

Looking ahead, I see clear opportunities to contribute to SIREN. My latent action primitives and physical inductive bias can directly strengthen SIREN's structured VLA components, while osmAG's semantic representation can serve as a prior for your grounded world models. I am particularly interested in exploring how these structured action representations can be integrated into action-conditioned world models to enable safer, more efficient long-horizon planning and human-in-the-loop adaptation — core elements of the holistic embodied intelligence SIREN seeks to realize.

I am eager to bring my systems-oriented mindset and industrial deployment experience to PEARL Lab. Under your guidance, I am confident I can help translate these ideas into real robotic systems that operate reliably in unstructured environments. I look forward to contributing to and growing within the SIREN project.

4. Degree Certificates and Academic Transcripts

Included degree certificates and academic transcripts

- Section 4.1 includes the official Master's enrollment certificate issued by ShanghaiTech University.
- Section 4.2 includes the Master's academic transcript issued by ShanghaiTech University.
- Section 4.3 includes the Bachelor's degree certificate.
- Section 4.4 includes the Bachelor's academic transcript.

上海科技大学
ShanghaiTech University

学籍证明

兹证明 张嘉杰 同学，性别 男 ，学号 2023233216 ，身份证号 330282200109047792 ，于 2023年09月 进入我校学习，系我校 信息科学与技术学院 全日制在籍 学术学位硕士 ，学制 3 年，预计于 2026年07月 毕业。

CERTIFICATION

This is to certify that Zhang Jiajie, male, born on September 04,2001, Student ID. 2023233216, ID card number.330282200109047792, has been studying the graduate program for Master's Degree in the School of Information Science and Technology, ShanghaiTech University, from Sept.2023 to now, with an expected graduation in July, 2026.

上海科技大学

2026年04月01日

www.shanghaitech.edu.cn
地址：上海市浦东新区华夏中路393号邮编：201210
Add:393 Middle Huaxia Rd.,Pudong,Shanghai 201210,China

上海科技大学
ShanghaiTech University

Transcript of Academic Record

Name: Zhang Jiajie **School of Information Science and Technology** **Date of Entrance:** Sep-2023

Student ID: 2023233216 **Program:** Master's Degree in Computer Science and Technology

TERMS	COURSE TITLE	CREDIT	GRADE	NOTE
Fall 2024	Deep Learning	4	B+	
	Academic Paper Writing	2	P	
	TERM TOTALS	**6**	**3.3**	
Spring 2024	Robotics	4	A+	
	Simultaneous Localization and Mapping	4	B+	
	Selected readings of classic works of Marx, Engels and Lenin	1	B+	
	TERM TOTALS	**9**	**3.61**	
Fall 2023	Algorithm Design and Analysis	4	B+	
	Computer Graphics II	4	B+	
	Principle and Practice of Robotic Manipulation	3	A	
	Comprehensive English I	2	P	
	Comprehensive English II	2	P	
	Research on the Theory and Practice of Socialism with Chinese Characteristics in the New Era	2	B	
	TERM TOTALS	**17**	**3.42**	

TRANSCRIPT TOTALS

	Degree required Credits	Earned Credits	GPA
TOTAL	**32**	**32**	**3.47**

上海科技大学 成绩证明专用章

Date Issued: 2025-09-06

Graduate School
www.shanghaitech.edu.cn
393 Middle Huaxia Road, Pudong, Shanghai 201210, China

上海科技大学
ShanghaiTech University

SHANGHAITECH UNIVERSITY
Graduate School
393 Middle Huaxia Road, Pudong District
Shanghai 201210, China

EXPLANATION OF TRANSCRIPT

Academic Year

Each Academic year includes 3 semesters: Fall, Spring and Summer. Fall and Spring Term have eighteen weeks, including exams, and Summer Term has four weeks.

Method of Assessment and Calculation of Scores

Examination results are recorded by letter grades or passing grades instead of percentage scores.

The conversion table for grade, grade point and corresponding percentage is as following:

Grade	A+	A	A-	B+	B	B-	C+	C	C-	F	P	NP	N
Grade Point	4.0	4.0	3.7	3.3	3.0	2.7	2.3	2.0	1.7	0.0	N/A	N/A	N/A
Corresponding Percentage	95-100	90-94	85-89	80-84	75-79	70-74	67-69	63-66	60-62	0-59	60	60	Record

P indicates pass; NP indicates Not Pass.

N indicates No Record, because the course is incomplete or the exam is postponed. When work completed, it will be replaced by final grade.

The method for calculating the GPA (Grade Point Averages) is:

GPA=(the course Credit×the course Grade Point) / the credits of all the courses taken

The course score P counts towards Credit, and both P and NP do not count towards the GPA.

If a failed course is retaken, only the retaken course's grade counts towards the GPA.

The special symbols' meanings

The course with a"▲" symbol is a retaken course.

The course with a"*" symbol is a buffer course.

The course with a"●" symbol is an out school course.

上海科技大学 成绩证明专用章

Zhengzhou University Degree Certificate

This is to certify that **Zhang Jiajie**, **male**, born in **September**, **2001**, has **majored** in **Automation**, has completed all the courses prescribed in the Teaching Program of **Undergraduate (Full Time)**, and has passed all the examinations necessary for graduation from **Zhengzhou University**. Having fulfilled the requirements stipulated by **Regulations of Zhengzhou University**, **he** is awarded **the degree** of **the Bachelor of Engineering** in **June, 2023**.

Certificate No.: **1045942023000498**

Operator	ShenJiao
Tel	0371-67763055

Archives and School History Museum of Zhengzhou University

April 2, 2026

Zhengzhou University Student Transcript

Student No:201902080230 College:School of Electrical and Information Engineering Study Duration:4Years

Name:Zhang Jiajie Major:Automation Class:2019020202

Course Name	Course Type	Credit	Grade	GP
1. 2019-2020 School Year First Semester Semester GPA: 3.06				
Linear Algebra	Obligatory	3	78	2.7
College English Reading and Writing (I)	Obligatory	2	92	4
Engineering Drawing	Obligatory	2	82	3.2
Physical Education (I)	Obligatory	1	90	4
College English Listening & Speaking (I)	Obligatory	1	92	4
Calculus A (I)	Obligatory	5	78	2.7
Introduction to Computer	Obligatory	2	82	3.2
Military Theory	Obligatory	1	60	1.2
Morality & Fundamentals of Law	Obligatory	3	83	3.2
2. 2019-2020 School Year Second Semester Semester GPA: 3.6				
College Physics A (I)	Obligatory	4	88	3.7
Care for Life-First Aid & Self-rescue Skills	Elective	2	83	
Calculus A (II)	Obligatory	5	90	4
College English Listening & Speaking (II)	Obligatory	1	90	4
Physical Education (II)	Obligatory	1	91	4
Probability Theory and Mathematical Statistics	Obligatory	3	84	3.2
College English Reading and Writing (II)	Obligatory	2	89	3.7
critical thinking (MOOC)	Elective	2	75	
High-level Programming Language	Obligatory	3	84	3.2
Etiquette for College Students	Obligatory	1	78	2.7
3. 2020-2021 School Year First Semester Semester GPA: 3.94				
Electric Circuits (Ⅰ)	Obligatory	3	92	4
Complex Function and Interal Transformation	Obligatory	2	95	4
Introduction to Maoism and Chinese Characteristic Socialism Theory System(I)	Obligatory	2.5	94	4
Electromagnetic Field	Obligatory	2	95	4
Digital Electronics	Obligatory	3	99	4
Information and Electronic Technology Altering the People's Life Style	Elective	2	82	
Outline of Chinese Modern History	Obligatory	3	87	3.7
College English (III)	Obligatory	2	92	4
Physical Education (III)	Obligatory	1	96	4
College Physical Experiment A (Ⅰ)	Obligatory	1.5	88	3.7
College Physics A (II)	Obligatory	3	90	4
4. 2020-2021 School Year Second Semester Semester GPA: 3.8				
Introduction to Automation	Elective	1	83	
Electric Machine & Drive	Obligatory	4	92	4
Electronics Comprehensive Technique Practice	Obligatory	1	B	3.7
Introduction to Maoism and Chinese Characteristic Socialism Theory System(II)	Obligatory	2.5	91	4
Electric Circuits (Ⅱ)	Obligatory	2	88	3.7
Electronic Experiment	Obligatory	1	B	3.7
Database	Elective	2	84	
Motor & Drive Experiment	Obligatory	1	A	4
Circuit Experiment	Obligatory	1.5	92	4
College English (IV)	Obligatory	2	87	3.7
Basic Principles of Marxism	Obligatory	3	88	3.7
Principles and Applications of Microcomputer	Obligatory	3	80	3.2
Analog Electronics	Obligatory	3	94	4
Numerical Computation & Matlab Programming	Elective	2	D	
College Physical Experiment A (II)	Obligatory	1.5	89.6	4
Physical Education (IV)	Obligatory	1	97	4
5. 2021-2022 School Year First Semester Semester GPA: 3.65				
An outline of Xi Jinping's thought of socialism with Chinese characteristics in the new era	Obligatory	2	88	3.7
Principle of Microcontroller	Elective	3	95	
Electronics Technique Practice	Obligatory	1	A	4
Metalworking Practice	Obligatory	3	83	3.2
Cognitive Practice	Obligatory	1	95	4
Engineering Practice of Artificial Intelligence	Elective	2	70	
Signal & System	Obligatory	3	84	3.2
Electronic Technology Curriculum Design	Obligatory	1	A	4
Detection & Conversion Technique	Obligatory	2	89	3.7
Special English	Elective	2	86	
Industrial Power Supply	Obligatory	2	88	3.7
Computer Networks	Elective	2	77	
Automatic Control Theory	Obligatory	4	92	4
6. 2021-2022 School Year Second Semester Semester GPA: 3.93				
Automatic Control Theory Curriculum Design	Obligatory	1	A	4
Modern Control Theory	Obligatory	2	97	4
Policy & Political Situation Analysis	Obligatory	2	92	4
Power Electronics	Obligatory	3	94	4
Detection & Process Control Experiment	Obligatory	0.5	A	4
Innovation and Entrepreneurship Education	Obligatory	2	89	3.7
Process Contro	Obligatory	2.5	93	4
Course Project of Power Electronics	Obligatory	1	B	3.7
Principle of Modern Communication	Elective	2	86	
7. 2022-2023 School Year First Semester Semester GPA: 3.68				
Technique of Computer Control	Obligatory	3	92	4
Course Project in Computer Control	Obligatory	1	80	3.2
Automatic Control System Experiment	Obligatory	0.5	Pass	3
Course Project in Automatic Control System	Obligatory	1	92	4
Intelligent control	Obligatory	3	84	3.2
Course Project of Intelligent Control	Obligatory	1	90	4
Career Planning and Career Guidance for College Students	Obligatory	2	90	4
Automatic Control System	Obligatory	2	86	3.7
8. 2022-2023 School Year Second Semester Semester GPA: 3.97				
Graduation Project	Obligatory	16	A	4
Production Practice	Obligatory	2	89	3.7
Automation Comprehensive Practice	Obligatory	2	90	4

Total Credits	178	Obligatory Courses Credits	156	Limited Courses Credits	0	Optional Courses Credits	22	Total GP	578.45	GPA	3.7	Average Grade	87.64

The Calculation of GPA doesn't include the elective course and the elective course doesn't calculate GP.
Scan code validation

学籍专用章

Printed by the Academic Affairs Office **Stamp of Academic Affairs Office** **Print Date:2023-06-26**

5. Academic References

Reference 1	Prof. Sören Schwertfeger
Title	Tenured Associate Professor & Co-Director of STAR Center, School of Information Science and Technology, ShanghaiTech University
Email	soerensch@shanghaitech.edu.cn
Relationship	Current Master's advisor at the MARS Lab, ShanghaiTech University
Reference 2	Dr. Alexander Kleiner
Title	Chief Expert in AI and Robotics & Director of Advanced Autonomous Systems (formerly at Bosch Corporate Research)
Email	alexander.kleiner@gmail.com
Relationship	Former supervisor during my internship at Wolong Electric (Jul. 2025 - Oct. 2025)